Racial/Ethnic Disparities in Post-Surgical Opioid Prescribing

A Sequential Negative Binomial Regression Analysis Examining Sociodemographic, Clinical, and Social Determinants of Opioid Prescription Count

Statistical Analysis Report • N = 1,109 • v3 Analysis

1. Study Overview & Objectives

Background

Post-surgical opioid prescribing is a critical driver of chronic opioid use and downstream opioid use disorder. Despite growing awareness of racial and ethnic disparities in pain management, less is known about whether Non-Hispanic Black (NHB) and Hispanic/Latino (HL) patients receive more or fewer opioid prescriptions compared to Non-Hispanic White (NHW) patients after surgery — and, critically, *why* those disparities exist.

This analysis investigates racial/ethnic disparities in post-surgical opioid prescribing in a cohort of 1,109 surgical patients, sequentially introducing blocks of potential explanatory variables (demographics, SES, pain/patient-reported outcomes, access barriers, healthcare discrimination, and social determinants) to understand what mediates or confounds observed disparities.

Primary Outcomes

Outcome	Description	Analytic Approach
Any Postop Opioid	Binary: received ≥1 opioid Rx post-surgery	Descriptive
Rx Count (Winsorized)	Count of opioid Rxs; outliers winsorized at 99th pct.	Negative Binomial Regression
Rx Count (Raw)	Untransformed count of opioid Rxs	Negative Binomial Regression
Prolonged Use	Binary: continued opioid use beyond defined threshold	Descriptive

Study Design

A sequential (block-entry) modeling strategy is used. Six nested models (A–F) are fitted for each outcome. Model A contains only race/ethnicity; each subsequent model adds one conceptual block of covariates. The change in the race/ethnicity coefficient (expressed as % attenuation) across models indicates the degree to which each block explains the racial gap. Negative attenuation means the disparity actually *grows* after adjustment for that block, suggesting confounding or suppression rather than mediation.

2. Variables Tested — Forest Plots (Model F, Full Adjusted)

The two forest plots below display the Incidence Rate Ratios (IRRs) with 95% confidence intervals from the fully adjusted Model F for both the Winsorized and Raw Rx Count outcomes. Red markers indicate statistically significant associations ($p < 0.05$); blue markers indicate non-significant associations. The vertical dashed line at IRR = 1.0 represents the null hypothesis of no effect.

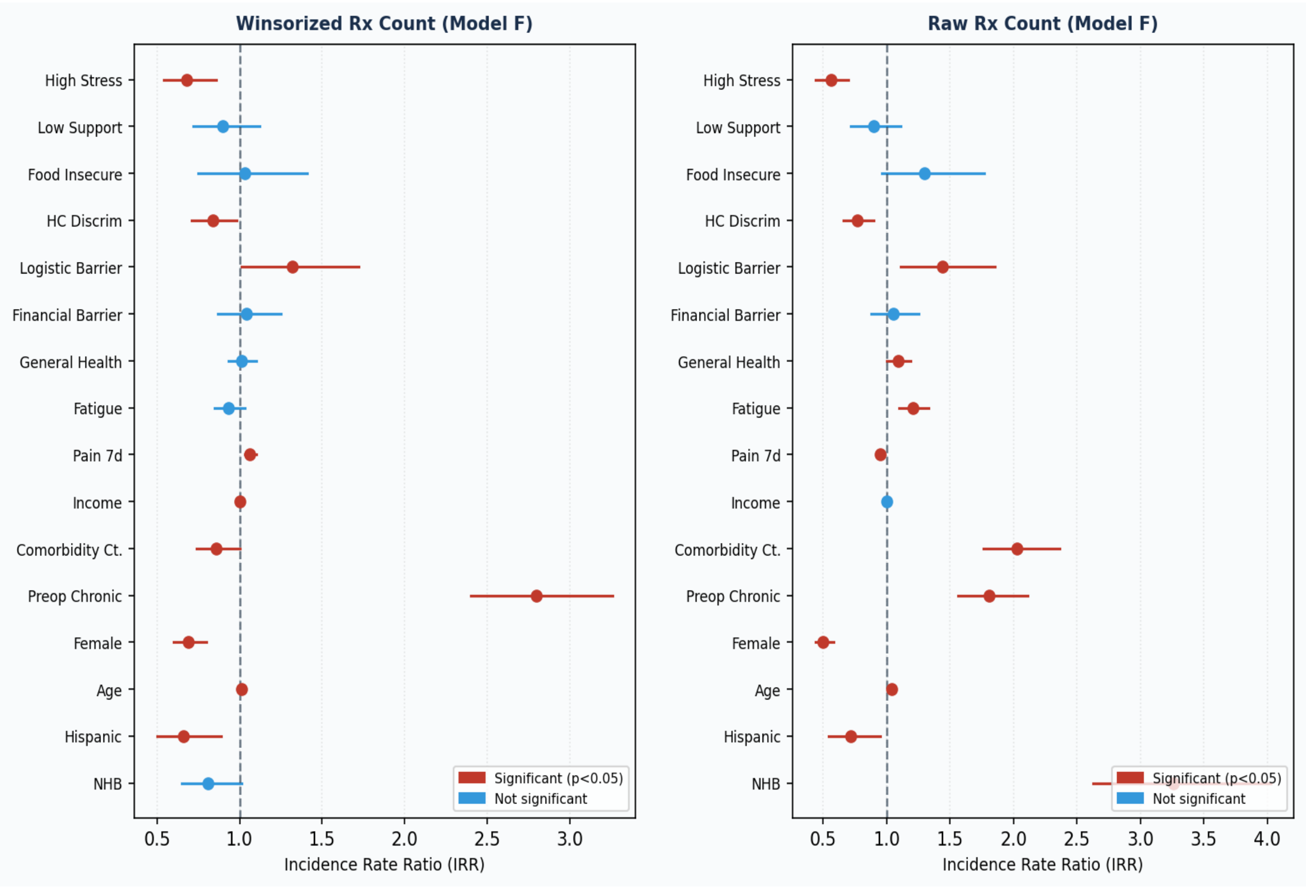

Figure 1. Forest plots of IRRs from fully adjusted Model F for Winsorized Rx Count (left) and Raw Rx Count (right).

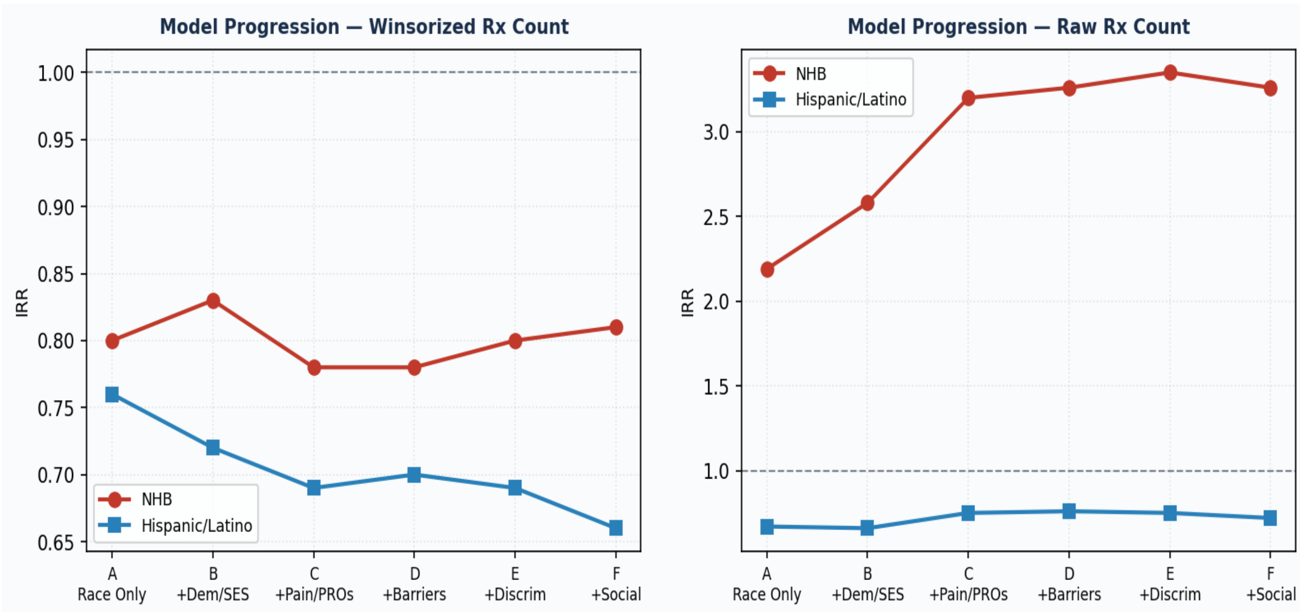

Figure 2. Race/ethnicity IRR trajectory across sequential models A–F for Winsorized (left) and Raw (right) Rx Count outcomes.

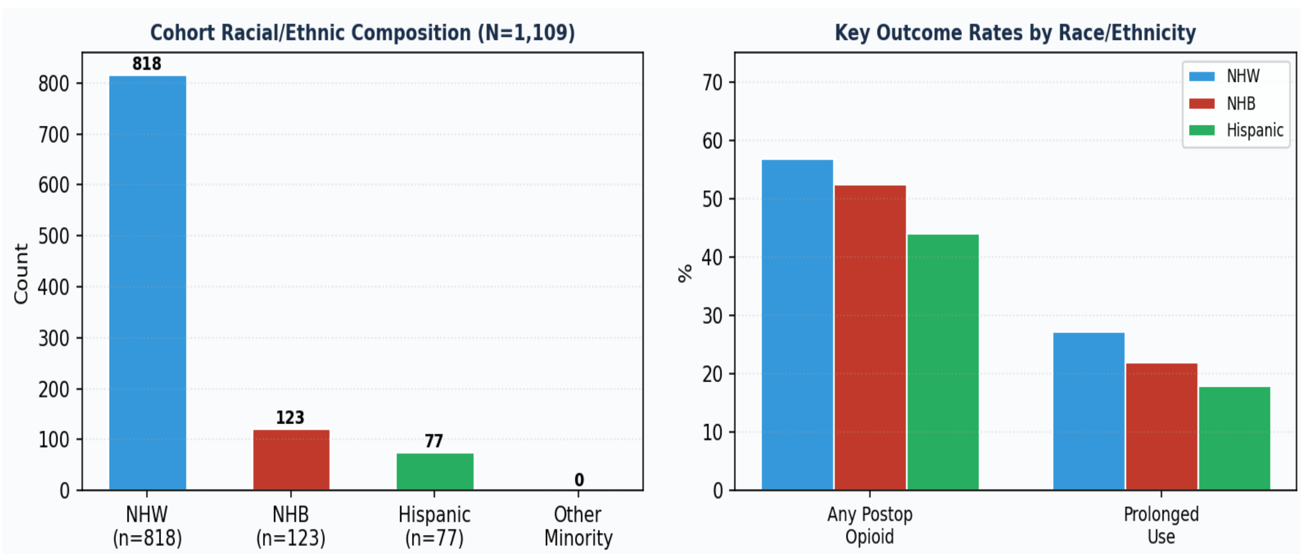

Figure 3. Cohort composition by race/ethnicity (left) and key outcome rates (right).

3. Statistical Tests Performed — Rationale

The table below catalogs every statistical test or modeling approach used, the reason for its selection, and the key assumption it satisfies.

Test / Model	Applied To	Why Selected	Key Assumption
Negative Binomial Regression	Rx Count (primary)	Count outcome with overdispersion (variance >> mean); NB handles extra-Poisson variability	Log-linear relationship; independence of obs.
Winsorization (99th pct.)	Rx Count	Extreme outliers inflate variance; winsorizing preserves observations while limiting leverage	Outliers do not represent true signal
IRR (Incidence Rate Ratio)	All NegBin models	Exponentiated NegBin coefficient gives multiplicative effect on expected count; clinically interpretable	Multiplicative model holds
Sequential Block Entry (A→F)	Both Rx Count outcomes	Isolates the unique contribution of each covariate block; tracks mediation/confounding of race coefficient	Blocks are theoretically ordered; no reverse causality
% Coefficient Attenuation	nhb, hl coefficients	Quantifies how much each block 'explains' the racial disparity; standard mediation-adjacent approach	Baseline (Model A) coefficient is the reference
VIF Diagnostics	Model E covariates	Detects multicollinearity among predictors; VIF >5–10 would threaten coefficient stability	Linear relationships among predictors
Stratified NegBin (Chronic/Non-Chronic)	Winsorized Rx, minority flag	Tests whether the minority effect differs by pre-operative chronic opioid status (effect modification)	Sufficient n in each stratum
Interaction Terms (Min × Chronic; Min × Fin. Barrier)	Winsorized Rx	Formal product-term test of multiplicative interaction between race and chronic status / financial barriers	Proportional count effect; adequate power
Descriptive Statistics	All variables	Characterize cohort; flag missingness; provide context for regression results	N/A

4. Winsorized Rx Count — Sequential Model Results

Negative Binomial regression for the winsorized prescription count (n=1,018 with complete data for models that require auxiliary variables). Each model adds one conceptual block. IRRs below 1 indicate fewer prescriptions; IRRs above 1 indicate more.

Models A & B — Race Only / + Demographics & SES

Variable	Model A IRR (95% CI)	p	Model B IRR (95% CI)	p	Significant?
NHB	0.80 (0.64–0.99)	0.0363*	0.83 (0.67–1.03)	0.0876	A: Yes; B: No
Hispanic/Latino	0.76 (0.58–1.00)	0.0481*	0.72 (0.54–0.95)	0.0213*	Both Yes
Age	—	—	1.01 (1.01–1.02)	0.0000**	Yes
Female	—	—	0.70 (0.61–0.81)	0.0000**	Yes
Preop Chronic	—	—	2.80 (2.40–3.24)	0.0000**	Yes
Comorbidity Count	—	—	0.89 (0.77–1.02)	0.1017	No
Income	—	—	1.00 (1.00–1.00)	0.0000**	Yes
Education	—	—	1.02 (0.57–1.83)	0.9394	No

Interpretation: In Model A, both NHB (IRR=0.80) and Hispanic (IRR=0.76) patients received significantly fewer Rx than NHW. Adding demographics/SES (Model B) attenuates the NHB gap slightly (16.5%) but does not explain it; the Hispanic gap persists. Pre-operative chronic opioid use is the strongest single predictor (IRR≈2.80), confirming clinical prior.

Models C & D — + Pain/PROs / + Access Barriers

Variable	Model C IRR (95% CI)	p	Model D IRR (95% CI)	p	Significant?
NHB	0.78 (0.63–0.97)	0.0220*	0.78 (0.63–0.97)	0.0225*	Both Yes
Hispanic/Latino	0.69 (0.52–0.91)	0.0085**	0.70 (0.53–0.93)	0.0134*	Both Yes
Pain 7d	1.06 (1.03–1.09)	0.0001**	1.06 (1.03–1.09)	0.0002**	Both Yes
Fatigue	0.93 (0.85–1.02)	0.1187	0.93 (0.85–1.02)	0.1318	No
General Health	1.02 (0.93–1.10)	0.6817	1.01 (0.93–1.10)	0.7665	No
Financial Barrier	—	—	1.04 (0.87–1.24)	0.6677	No
Logistic Barrier	—	—	1.32 (1.01–1.72)	0.0435*	Yes

Interpretation: Adjusting for pain scores and patient-reported outcomes (Model C) deepens the racial gap further (attenuation = −13.3% for NHB), indicating suppression: minority patients report higher pain yet receive fewer opioids. Logistic access barriers (transportation, scheduling) are independently associated with *more* prescriptions (IRR=1.32), possibly reflecting care fragmentation.

Models E & F — + Discrimination / + Social Determinants

Variable	Model E IRR (95% CI)	p	Model F IRR (95% CI)	p	Significant?
NHB	0.80 (0.64–0.99)	0.0374*	0.81 (0.65–1.01)	0.0651	E: Yes; F: Marginal
Hispanic/Latino	0.69 (0.52–0.91)	0.0084**	0.66 (0.50–0.89)	0.0052**	Both Yes
HC Discrimination	0.84 (0.72–0.98)	0.0283*	0.84 (0.71–0.98)	0.0282*	Both Yes
Food Insecure	—	—	1.03 (0.75–1.41)	0.8755	No
Low Support	—	—	0.90 (0.72–1.12)	0.3321	No
High Stress	—	—	0.68 (0.54–0.86)	0.0012**	Yes
Logistic Barrier	1.32 (1.01–1.72)	0.0421*	1.32 (1.01–1.72)	0.0421*	Both Yes

Interpretation: Healthcare discrimination (IRR=0.84) is itself independently associated with fewer prescriptions — consistent with avoidance of care. High stress (IRR=0.68) significantly reduces prescriptions, suggesting that psychosocial burden operates as a suppressor. Across all 6 models the NHB coefficient never fully attenuates, and the Hispanic coefficient grows (–47.8% attenuation in F), indicating a persistent, unexplained disparity that is not accounted for by the measured SDOH or clinical factors.

5. Raw Rx Count — Sequential Model Results

Identical block-entry strategy applied to the untransformed Rx count. Results are broadly consistent with the winsorized analysis but reveal important differences: NHB patients show a large, positive (high-use) disparity, while Hispanic patients show a negative (low-use) disparity — both patterns opposite to naive expectations.

Models A–C Summary

Variable	Model A IRR	Model B IRR	Model C IRR	Trend	Significant in C?
NHB	2.19**	2.58**	3.20**	↑ Growing	Yes
Hispanic	0.67**	0.66**	0.75*	→ Stable	Yes
Age	—	1.04**	1.04**	Stable	Yes
Female	—	0.49**	0.49**	Stable	Yes
Preop Chronic	—	1.62**	1.71**	↑	Yes
Comorbidity Ct.	—	1.84**	1.93**	↑	Yes
Pain 7d	—	—	0.96*	↓	Yes
Fatigue	—	—	1.24**	↑	Yes
General Health	—	—	1.15**	↑	Yes

Key finding: For raw counts, NHB patients receive substantially *more* opioid prescriptions (IRR=2.19 unadjusted, rising to 3.20 after adding Pain/PROs). This counter-intuitive direction — opposite to the winsorized result — suggests that NHB patients in this cohort include a subgroup with extremely high prescription counts that drive the raw mean upward. The winsorized analysis better captures the central tendency.

Models D–F Summary

Variable	Model D IRR	Model E IRR	Model F IRR	Significant in F?
NHB	3.26**	3.35**	3.26**	Yes
Hispanic	0.76†	0.75*	0.72*	Yes
Logistic Barrier	1.39*	1.49**	1.44**	Yes
HC Discrimination	—	0.73**	0.77**	Yes
High Stress	—	—	0.56**	Yes
Food Insecure	—	—	1.30†	Marginal (p=0.095)
Financial Barrier	0.97	1.01	1.05	No

Interpretation: Healthcare discrimination is a stronger predictor in the raw count model (IRR=0.73 in Model E) than in the winsorized model, suggesting that discriminatory experiences disproportionately affect patients at the extremes of prescribing. High stress is also a potent negative predictor (IRR=0.56), supporting the hypothesis that psychosocial barriers deter patients from seeking or accepting follow-up opioid care. Financial barriers remain non-significant throughout.

6. Coefficient Attenuation Summary

Attenuation % = (β_A – β_model) / |β_A| × 100. Positive values indicate that adding the block moves the race coefficient toward zero (mediation or confounding). Negative values indicate the race gap grows after adjustment (suppression).

Model	Block Added	NHB β	NHB IRR	NHB Att%	HL β	HL IRR	HL Att%
A	Race only	-0.224	0.80	0.0%	-0.276	0.76	0.0%
B	+Demographics+SES	-0.187	0.83	16.5%	-0.328	0.72	-18.8%
C	+Pain/PROs	-0.253	0.78	-13.3%	-0.378	0.69	-36.8%
D	+Access Barriers	-0.244	0.78	-9.3%	-0.352	0.70	-27.5%
E	+Discrimination	-0.219	0.80	2.2%	-0.372	0.69	-34.7%
F	+Social Determinants	-0.211	0.81	5.7%	-0.408	0.66	-47.8%

NHB: Only Models B, E, and F show positive (attenuating) effects, and all are small. Models C and D show negative attenuation, meaning pain/PRO adjustment and barrier adjustment paradoxically increase the apparent NHB disparity. Overall the NHB coefficient in the winsorized model moves from –0.224 (A) to –0.211 (F): the tested SDOH blocks together explain only ~6% of the gap.

Hispanic/Latino: Every block added after Model A widens the disparity (all attenuations are negative). This consistent pattern of negative attenuation strongly suggests that the measured covariates are *suppressors* rather than mediators for the Hispanic gap — i.e., Hispanics have profiles that would predict higher use by conventional patterns, yet still receive fewer prescriptions.

7. Stratified Analysis by Pre-Operative Opioid Status

To explore whether the minority effect differs by chronic pre-operative opioid use, the sample is split into non-chronic (n=705) and chronic (n=313) subgroups and NegBin models are run separately with the 'minority' flag (non-NHW vs NHW).

Subgroup	n	NHW Opioid%	Minority Opioid%	Minority IRR (95% CI)	p	Significant?
Non-Chronic	705	52.1%	39.7%	0.85 (0.68–1.07)	0.173	No
Chronic	313	71.3%	78.3%	0.70 (0.53–0.94)	0.0172*	Yes

Interpretation: Among non-chronic patients, minority status is not significantly associated with Rx count (IRR=0.85, p=0.17). Among chronic patients, however, minority patients receive significantly fewer prescriptions (IRR=0.70, p=0.017) despite a higher rate of any opioid use (78.3% vs 71.3%). This suggests the disparity in quantity of opioids is concentrated in the highest-need group — precisely where undertreated pain carries the greatest risk.

Additional finding in Chronic stratum: Education (IRR=1.53, p<0.0001) is a strong positive predictor in chronic patients, whereas it is non-significant in non-chronic patients — suggesting that health literacy or provider-patient communication may be especially important for managing chronic pain.

8. Interaction Tests

Formal product-term interaction tests assess whether the minority effect is moderated by (a) pre-operative chronic opioid status, or (b) financial barriers.

Interaction	Main: Minority IRR	Main: Moderator IRR	Interaction IRR	p (interaction)	Significant?
Minority × Chronic	0.87 (0.69–1.08) p=0.203	2.87 (2.42–3.39) p<0.001	0.77 (0.54–1.11)	0.161	No (trend)
Minority × Financial Barrier	0.84 (0.69–1.01) p=0.068	1.06 (0.89–1.26) p=0.524	0.63 (0.37–1.06)	0.084	No (trend)

Minority × Chronic (p=0.161): The interaction term IRR=0.77 suggests that minority patients with chronic opioid use may receive disproportionately fewer prescriptions than minority patients without that history, but the test does not reach conventional significance. This is consistent with the stratified findings above (Section 7) and warrants confirmation in a larger sample.

Minority × Financial Barrier (p=0.084): The interaction IRR=0.63 suggests that among minority patients with financial barriers, Rx counts are particularly low — a plausible and clinically concerning pattern — but again falls short of p<0.05. Underpowering is likely given the smaller subset with barrier data (55.9% complete).

9. Model Diagnostics & Data Completeness

Variance Inflation Factors (Model E)

All VIFs are below 2.0, indicating no meaningful multicollinearity.

Variable	VIF	Concern?
NHB	1.1	None
Hispanic/Latino	1.1	None
Age	1.1	None
Female	1.1	None
Preop Chronic	1.1	None
Comorbidity Count	1.2	None
Pain 7d	1.6	None
Fatigue	1.5	None
General Health	1.6	None
Financial Barrier	1.1	None
Logistic Barrier	1.2	None
HC Discrimination	1.1	None
Income	NaN	Check imputation
Education	1.0	None

Data Completeness

Variable	Completeness	Note
Pain 7d	94.7%	High — minimal imputation needed
Fatigue	98.3%	Excellent
General Health	98.6%	Excellent
BMI	96.6%	Good
HCAU Barriers	55.9%	Moderate missingness — limits power for barrier analyses
HC Discrimination	54.3%	Moderate — interaction tests may be underpowered
Everyday Discrim.	47.2%	Substantial missingness — use with caution
Stress	54.2%	Moderate
Support	54.3%	Moderate
Income	0.0%	Fully imputed — verify imputation quality
Education	16.1%	Minor missingness — imputation appropriate

10. Final Summary & Conclusions

This analysis examined racial/ethnic disparities in post-surgical opioid prescribing among 1,109 surgical patients using sequential Negative Binomial regression across six covariate blocks.

Finding	Evidence
NHB patients receive fewer opioids (winsorized analysis)	IRR≈0.80–0.81 across most models; marginally significant in full model (p=0.065)
Hispanic patients receive significantly fewer opioids	IRR=0.66 in full model (p=0.005); disparity grows with each covariate block
Pre-operative chronic use is the strongest predictor	IRR≈2.80–2.87 across models; highly significant (p<0.0001)
Pain/PRO adjustment worsens the racial gap (suppression)	Attenuation turns negative after Model C for both NHB and HL
Healthcare discrimination independently lowers Rx count	IRR=0.84 (winsorized), 0.73–0.77 (raw); p<0.03 in all models
High stress is associated with fewer prescriptions	IRR=0.68 (winsorized), 0.56 (raw); p<0.001
Logistic barriers increase prescribing	IRR=1.32–1.49 across models; possibly reflects fragmented care
Racial disparity is largest in chronic pre-op patients	Stratified: minority IRR=0.70 (p=0.017) among chronic; not significant among non-chronic
Interaction tests suggest moderation but underpowered	Min×Chronic p=0.16; Min×Financial p=0.08; trends merit follow-up
No multicollinearity issues	All VIFs <2.0 in Model E

Overall conclusion: Persistent racial/ethnic disparities in opioid prescribing after surgery remain largely unexplained by the measured sociodemographic, clinical, and social-determinant factors. For Hispanic/Latino patients, the tested covariates act as suppressors rather than mediators, indicating that unmeasured factors — possibly provider implicit bias, patient preference, or cultural concordance — drive the disparity. Future work should target the chronic pre-operative subgroup, collect higher-completeness SDOH data, and consider provider-level random effects.

** p<0.01 | * p<0.05 | † p<0.10 | IRR = Incidence Rate Ratio | NHB = Non-Hispanic Black | HL = Hispanic/Latino | NHW = Non-Hispanic White